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Timbre and composition – timbre and language¹

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The function of timbre in 20th Century instrumental music is discussed in terms of the relation between timbre and musical language. Up to the 19th Century, the function of timbre was primarily related to its identity in addition to being charged with certain effective and symbolic characteristics. The identities of Western instruments are standardized, investing them with a certain neutrality that allows the construction of pitch hierarchies that are unperturbed by differences in timbre of the instruments. For small ensembles, timbre has a stability and a separating power that provides an identification and clarification of form and timbre. The function of timbral identity is one of articulation. With the modern orchestra, the use of instruments is more flexible and their identification becomes more mobile and temporary. Composing the blending of timbres into complex sound-objects follows from the confronting of established sound hierarchies and an enriching of the sound vocabulary. The function of timbre in composed sound-objects is one of illusion and is based upon the technique of fusion. One contrasts, then, notions of raw timbre and organized timbre. The importance of discontinuity in musical dimensions for composition is considered with respect to timbre. Finally, the relation between timbre, composition and sound transmission in an acoustic space is discussed.

KEY WORDS timbre, composition, musical language, articulation, identity, fusion, illusion.

The subject of my paper was announced as being the function of timbre in 20th Century instrumental music. In fact I would prefer to deal with the relation between timbre and musical composition (*écriture*²) and more generally speaking, timbre and language. Whatever the intention, it can be said that there are basically two ways of considering timbre: one is an objective, scientific way, beyond language, and without aesthetic criteria, in which there is an extreme difficulty in moving from the quantitative aspect back up to the qualitative. With the help of graphs and diagrams many acoustic phenomena can be described but the quality of integration of sound and timbre in the structure of a composition is absent from these measurements. Even when one deals with the perception of sound phenomena and their quality, it is mostly a question of perception in isolation, exempt from any context. I feel that the truly artistic value of timbre is fundamentally forgotten using this approach.

On the other hand we have the subjective, artistic manner of dealing with timbre, as a constituent of a musical language, along with the aesthetic and formal criteria which relate to it. This leads us to an opposite difficulty, namely the impossibility of linking instinctive feelings about the qualitative aspect to a more reasoned appraisal of the quantitative. This difficulty becomes evident when composers with a traditional musical education based on instrumental knowledge are confronted with sound synthesis. They find themselves out of their element because the notion of quantity which is essential to the organization of synthetic sound is completely foreign to them. They are no longer dealing with established categories. Musicians in general are not interested in measurement or objective analysis. What matters to them is the function of timbre as related to composition, and, even more so, the affectivity created by the perception of timbre in the context of the work.

One can object to this by saying that the timbre of an instrument is defined by the manufacturer without reference to any stylistic criteria. If this had not been the case, many of our instruments would never have survived such an enormous evolution of styles. If the violin had only been associated with baroque music, it would obviously not have survived. It possessed, however, infinitely more stylistic possibilities than would have ever been dreamt of by its original makers. A clarinet manufacturer doesn't ask himself questions about the ultimate use of the instrument he makes. He creates an object which can serve equally well for a country choral society as for the interpretation of a highly elaborate piece of music. The instrument maker is in principle neutral, although not totally: instruments are made to follow a precise musical system and a fairly rigid standardization.

When one wants to go beyond certain types of instrumental usage, one comes up against a hierarchy embodied in the instrument and one risks creating marginal and anecdotal effects. Violin tuning is based on the 5th, an interval essential in the tonal system, with no particular importance in the 12-tone system. Violin positions follow a certain hierarchical conception of the universe of intervals, and the further one moves away from this, the more the playing of the instrument becomes difficult. In the same way, if one refuses the primacy of the semitone on the clarinet, one will come up against difficulties, having to avoid orthodox solutions by using special fingerings and alternative embouchures. Even though circumscribed by standardization, the instrument can escape the control of the system by the use of "peripheral" techniques.

To understand the extent to which timbre, composition and affectivity are linked in the mind of the composer, one only needs to look at the musical education he has received, and which he himself transmits. Instrumentation is not learned by a systematic study of timbre, but by picking out here and there examples, chosen as models, which work particularly well. All the time that is necessary is taken, and sometimes more, in the study of the laws of harmony and their evolution. In the

same way, counterpoint is studied at great length, various aspects being covered in extensive detail. Musical form is also taught: the fugue for example, a very coherent and powerful form, where a compositional technique is developed from the study of historical models, obeying rules which are much stricter than those used in the originals themselves.

But how far does the study of instrumentation go? Only as far as descriptions of a practical nature and methods more or less derived from examples divorced from their contexts. These practical descriptions can be clearly seen in a book on orchestration. One can look up one instrument after another, read a description of their registers with respect to range, dynamic levels, articulation, speed, breathing limits, etc. . . . Beyond this are found some recipes taken from the repertoire: disparate, incoherent examples to which are linked certain specific emotional effects.

Berlioz' treatise remains the prototype of this genre, because it is the most intelligently written – sharp and astute, completely and accurately reflecting his views on the summation of orchestration – a treatise which has survived extremely well. In it one finds an intimate mixture of the practical, the emotional and the symbolic, characteristic of the attitude of the musician. Berlioz associates the French horn with Weber's Oberon, the timbre of enchantment. He tells you, of course, that the lowest register of the clarinet sounds "hollow," but quickly adds mention of anguish, and sombre feelings. He notes the brilliance of clarinets in the middle and upper-middle registers, going on to make an analogy with the voices of women cheering warriors into battle. Today these expressions are amusing. Nonetheless they show an approach linked closely with symbolism and affectivity. Such an interpretation is not merely limited to timbre, it also applies to different keys: the heroic key of Eb, the pastoral key of F. The shadow of Beethoven hangs over these too precise associations, derived directly from well-loved examples and chosen with this catalogue of tonalities in mind. When musical language left behind such strongly coded hierarchies, these connotations more or less disappeared. It would nowadays be extremely difficult to keep to a point of view so closely linked to masterpieces from a given period.

It is not, however, the only characteristic of our evolution that timbre has lost the exclusive value of its function as a "sign-post for the emotions." I would like to emphasize the fact that, previously, musical language was based on the recognizable identities of its constituent elements. It comes as no surprise to note that the orchestra has developed in the direction of standardization: by ordering the means of sound production, such as grammar orders language. At the high point of tonality this identity was fundamental: the note D is above all a D. Whether it is played on a violin, a clarinet, or a trumpet, it must be recognizable as being the note D. In this way our instruments were built to facilitate and confirm this hierarchy. This standardization has in a sense impoverished the family of timbres, but it has enabled communication between them. On the other hand, one has noticed that in other

civilizations each instrument, even each part of an instrument, is endowed with a separate power: each sound in its struggle against standardization is allocated a range and at the same time a timbre, or a certain auxiliary quality which gives the sound a deeply individual character. As far as our civilization has, in a sense, favored the neutrality of the elements included in a general hierarchy, so other civilizations have given priority to the individual quality of these elements. Today, as there is no longer a harmonic language based on a general hierarchy, the note D is only what it is: an ingredient of a chord. There is nothing essential about its identity. On the contrary, the greater the malleability of the note D, the more it will lend itself to differential organizational methods, and the greater the satisfaction it will give us. This transformation will be reflected in the timbre, no longer considered as a principle of identification but as a principle of transition, if not of confusion.

In Baroque music (the Cantatas, Passions or the Brandenburg Concertos), the instrumental group is small, clearly perceptible as a group of individual instruments. Moreover it remains fixed in a piece, or a group of pieces. There is not yet a sense of an orchestra as we understand this notion: none of its neutrality or generality. Thanks to a given instrumental group, timbre helps one to perceive the totality of the work. An aria with two *cors anglais*, viola da gamba, flute or solo violin will keep its characteristic sound for most of the piece, by means of its clearly identifiable timbre. The instrumental timbre has stability and a single function: it defines a world which refers uniquely to itself, and provides a total identification of the form with the timbre. As one approaches the definition of the orchestra as we understand it, instrumental entities become more diffuse, the identification of instruments becomes mobile and temporary, the specific emotional effect associated with a particular instrument becomes less evident.

It is generally accepted that the modern orchestra was a creation of the 19th century. Effectively it was born as a result of a flexible use of instruments: timbre was to model itself upon various aspects of form. With the growing size of the orchestra, the role of the instrument becomes not blurred but flexible, multifarious, and correspondingly the forms expand. Even in large scale Baroque works one sees an accumulation of relatively small scale forms. In the period when the orchestra developed, from Beethoven to Mahler, forms increased in scale, based on transitions and multiplicity. In the same way, timbre took on a multiplicity of uses and instrumental characteristics. In certain usually relatively short passages an instrument would be used in a soloistic capacity, to be immediately identified with a given expression. The opposite happens at other points: the group dominates. The woodwind or brass ensembles, or more frequently mixtures between groups, create temporary ensembles. As soon as this happens the instruments take on a neutral quality. Their role evolves from the identifiable to the unrecognizable, because of the complexity and brevity of their blending. Indeed it is an understatement to say that the principle of their individual identity is brought into question! The instrument is

exploited for something beyond its individual qualities: its potential for fusion, for being neutral, for losing its individual quality, the latter quality naturally impeding the phenomenon of fusion.

From Schönberg onwards instruments become increasingly considered as being part of a texture, or constituent elements of varied textures, each time appearing in a different context. A typical example of this approach, and in a sense the archetype, is the third of the Opus 16 pieces, called *Farben* (colors in German), where the rhythmic, motivic and harmonic components are extremely rarefied. Only timbre with its multiple identities creates cohesion between the different elements.

The identity of sound is no longer seen as being the basis of the language, but instead this identity is progressively created from the needs of the language, is created *by* these needs themselves. As a result the orchestra has increasingly admitted instruments which don't conform to the accepted dominant hierarchy, instruments which are completely alien to it.

In the main, instruments which have been imported or adopted from other civilizations have been introduced, with the idea that they would easily integrate. The percussion section, for example, shows the most visible recent transformation of the body of the orchestra, admitted at first as a sporadic, picturesque element restricted to a few instruments. It has grown to an extensive number of instruments from cultures foreign to the "classical" orchestra. Gradually it became an important constituent of the collection of orchestral sounds; but these instruments don't obey the hierarchy to which the others belong and so a certain number of them is necessary to create another hierarchy based essentially on timbre. It has been forgotten that some of these imported instruments belong to very precise hierarchies. If they are not abstracted from the civilization to which they originally belonged, they will keep with them disparities of culture which will create problems for the unity of the work. They can enrich the sound world, but only as an "exotic" adjunct. In the same way, the marginal use of traditional instruments destroys the hierarchy to which they originally conformed, without the addition of anything other than a peripheral effect. In the orchestral sphere one can see the desire to create sound objects from traditional instruments which are held together by a hierarchy other than that which governed the making of these instruments. At the same time one observes the intrusion of ancillary phenomena into the orchestral world, calling into question by their very existence the hierarchy in which they are made to participate.

The slow evolution of the musician's sound-world, slow because it runs up against much important acquired experience, is certainly due to the evolution of musical thought with respect to organizational hierarchies. This thought is increasingly dependent on the work itself, at the moment of its creation, and is thus reduced to temporary and provisional states. Endeavors, whether in the vocal or instrumental fields, have been aimed at confronting the established hierarchy, at rejecting a standardized order.

At this point I would like to make a suggestion concerning vocal music. Even though we note an important evolution in the 19th century of what is correctly called the modern orchestra, we can observe, in vocal music on the other hand, a near immobility. Evolution in style, yes; in the intervals used, yes; in the harmonic vocabulary, yes; but vocal writing has hardly changed to this day. Only very recently has vocal delivery become diversified. In the most innovative works, vocal writing remains traditional. Schönberg, in *Die Glückliche Hand*, in *Moses and Aaron* especially, mixed different types of delivery, spoken and sung — in proportions which disturb the hierarchy, because the problem of this technique hasn't been stated clearly enough. For example the rhythmic structures used for singing and spoken passages are identical. When voices are superimposed, a sort of harmonic mud is the result, because in sung delivery pitches are selected, limited and circumscribed, whereas in spoken delivery they are anarchic and beyond control. The result is effectively a dirty environment, so to speak, surrounding the phenomenon of song which is very pure. Cohesion here becomes a problem because the spoken delivery disturbs the singing, without the latter orienting the former. The problem is incompletely treated, but it is at least tackled. It is examined more closely in this case than in *Sprechgesang* where the conception arises, in my view, from an erroneous analysis. For over three quarters of a century a satisfactory solution to the problem posed in *Pierrot Lunaire* has still not been found. Only lame solutions have emerged, referring to one factor more than another, preferring one mode of delivery over another. I close this digression by observing that the evolution of the use of vocal timbre has come much later than the evolution of instrumental timbre. I cannot propose a rational explanation for this, but I think that for many reasons, some of which are non-musical, vocal culture has been more important than instrumental culture in the evolution of music up to a certain point in its history. Vocal music has remained rooted in the vocal tradition, which preceded the instrumental tradition and survived it for a certain period.

Let us return to the notion of timbre, whether instrumental or vocal. Timbre exists aesthetically when it is directly bound to the constitution of the musical object. On its own timbre is nothing, like a sound on its own is nothing. Obviously a sound has an identity; but this identity is not yet an aesthetic phenomenon. Aesthetic identity only appears if there is utilization, language and composition. Unless one has arrived at this stage, objects exist by themselves, available, but empty of meaning. In the same way, a spot of color is definable as being blue, or red; but it does not induce in us any sense of a pictorial world.

Composition can start from within objects, in order to build them, or from the outside, in order to organize them. These are two types or two stages of composition. Composition from without puts these different objects together in a formal context, with a view to development. External criteria can act on internal criteria and modify the objects in order to link them in a coherent development and place them in a formal

context. Or the objects can be relatively neutral, organized in a simple way from within, even reduced to a single component. Then the composition of relationships becomes paramount, and very complex. Or you can organize complex objects from the inside, but you can only manipulate them in a relatively simple manner. Complexity of organization, of composition, can move from the inside to the outside, but in my opinion it cannot exist on the same level in both cases. Moreover one has to consider that in the area of sound synthesis, as in the purely instrumental realm, complex objects have a tendency not to be neutral with respect to a formal context. A complex object possesses a centrifugal force, making it capable of entering into a conflict with the context since it retains its identity. Its strength of identity tends towards an autonomy in relation to the context, and can go as far as destroying it. A simple, malleable object can, on the contrary, progressively modify its identity according to the event in which it participates. When the principle of identity, of identification, is too strongly reinforced from within, it becomes of little use without, and vice versa. All methods are legitimate, provided one accounts for their specificity.

Firstly as a model I should like to take the world of instrumental music since it provides many examples, taken from years of experience. One can then project conclusions on the future by extrapolation. The world of instrumental music is effectively divided into two. This division subsists despite many intermediate marginal areas. I will call them the world of small ensembles, or chamber music, and the world of large groups, or orchestral music. Each of these worlds, especially at their extremes, opens up a number of possibilities which are, in general, mutually exclusive, in spite of certain marginal interferences.

The small ensemble, primarily uses the analysis of discourse by means of timbre, creating interest by refinement and division, while the large ensemble primarily uses multiplication, superimposition, accumulation, creating an illusion, what Adorno called (in another context) phantasmagoria. The large ensemble, the orchestra, is the model, even, of the instrument of illusion, of phantasm, while the small ensemble represents the world of immediate reality and analysis. The small ensemble is preferably the world of articulation, while the large ensemble is essentially the world of fusion. Articulation and fusion, these are the opposite poles of the use of timbre in the instrumental world.

A characteristic example of the small ensemble is the *Klangfarbenmelodie* as found in the Webern Opus 10 pieces. In the first piece in particular the melodic line is analyzed by timbre, each note being vested in an instrumental color, each articulation underlined by a change in timbre. The problem of melodic understanding through timbre is fascinating. When played on a single instrument the melodic line has an immediately clear continuity, but you have to conceive your own articulation. Your comprehension, your participation, or simply your habit of mind cause this melodic line, with the help of the performer who highlights the phrasing in playing it, to appear first in its continuity, and then in its articulation. If I want to demonstrate this articulation through timbre,

like Webern I will distribute the timbres according to the articulation of the phrase. But in doing this I have introduced an element of diversion, of difference in timbre, which breaks the continuity so evident in a phrase with a single timbre. I have, at the same time, created a perceptual difficulty. The more one wants to produce clarity, the more one risks ending up with obscurity. At a certain point, things go the other way: the more you explain the fundamental construction of a phrase by timbre, the more you make its totality difficult to perceive, because you have mixed different categories, which have a tendency to take their autonomy and tear up the continuity which, on the contrary, you wanted to preserve by over-explaining it.

Starting from this point one can play with the multiplicity of timbre in relation to, amongst other things, the unity of the melodic line, in other words to play on the identification or the impossibility of identification. To illustrate this I shall take an example from the *Symphony* Opus 21 by Webern. The exposition in the first movement is a double canon. The main canon is identifiable by means of a clear distribution of timbres — the parts of phrases are sufficiently long and explicit with a fairly regular change of timbre. The secondary canon is more discontinuous in its phrasing, as it is in its distribution of timbres. As such it is difficult to identify. One sees the wish to contrast a phrase which is less analyzed and more clear with a phrase that is more analyzed and less clear, resulting in the distinct differentiation between the two levels of this double canon.

One can also use the brevity of a sound object or its length to orient or disorient the perception of timbre. In my work *Eclat*, certain chords are played simultaneously by resonating instruments. The different timbres which make up this chord would be recognized if I were to leave to each instrument its individual duration of resonance. If all the instruments play the chord staccato, then I suppress this natural means of analysis, and our perception can no longer discern which combination is in use in the block of sound. Identification depends in this case on the presence or absence of elements essential to perception.

Large ensemble technique is very different. My first example concerns the illusion of timbre which can be created by harmony. Below a certain level of perception, that is to say beyond a certain speed, a succession of chords will be perceived as a mixture of timbres rather than as a superimposition of pitches. Effectively these chords do not obey accepted harmonic functions and our perception is unable to analyze the phenomenon of these fleeting pitches. The vertical intervallic relationships are there to create a sound-object and not to establish functional relations. If such a chord is attached to a horizontal line, without any internal modifications, its identity is absorbed by our perception: as a timbre-object the chord *thickens* the line. If you vary the articulation or the relationship between the intervals, or if the different instrumental components change across the structure of the chord-object, the illusion of a created timbre in continuous evolution is evidently much stronger, our perception having been pulled in different directions by equal

forces. One will have created the opposition between the hidden constant and an apparent diversity.

The beginning of *Eine Blasse Wäscherin* in *Pierrot Lunaire* is a prototypical example of this interplay of melody, harmony and timbre: only three instruments – violin, clarinet and flute – follow each other in playing the top note of a series of three-note chords to form a melody of timbres which emerges from the harmony. This is a small ensemble example, but the technique used is that of illusion. The technique of accumulation is typical of the large ensemble: one finds it especially in late Debussy, but it already exists in Wagner. A single sound universe is transcribed by the use of different timbres into rhythms and figurations which are slightly divergent: clarinet tremolos doubled by string pizzicati, for example. One reality seen through slightly divergent prisms. The superimposition of two or more of these prisms will create the illusion of a complex timbre: a coagulation, a fusion which emerges from the way the passage is composed. This is what matters. Timbre does not function on its own, but the acoustic illusion of timbre is brought out by the way the music is composed. From this I deduce two notions of timbre used in instrumental music: raw timbre and organized timbre. In the first case composition acts from the outside. In the second case it works from the inside of the sound-object. The reality and identity of the instrument can thus be enveloped in a network of ambiguities which either hides it within a fused sound-object or reveals it in its absolute state. In this case timbre is related to the evolution of musical language and contributes to its enrichment.

One must not forget that the very realistic factors of time and space – involving transition between sounds, emission of sounds, the actual space separating instruments – can present serious obstacles to the realization of these illusions. That is why it is so important to plan the placing of instruments, the stage setting and the acoustics, at the very time the work is being composed, as composition and timbre are essentially confirmed by the appropriate acoustic arrangement. Thus we are going against the standardization reached at the beginning of the 20th Century. From now on timbre, composition and acoustic setting should be linked by the same necessity, unique to the work in question.

The identification of raw or organized timbres is to be found in an area between the extremes of immediate perception and elusive perception, with the unique possibility of playing on memory – the memory of an event, a sound object you have mentally recorded without having had the time to analyze it. For example, you may partially remember, thanks to one component, a striking event. In this way you give your memory the possibility of analyzing or perceiving retro-actively this component from the sound-object which has been heard. This virtual memory of the heard object constitutes one of the most important and interesting phenomena in composition, from the moment it can be integrated into the development of form and thematic material. The memory's comings and goings from the past to the present, its possible projections towards the future, give the work a range of perspectives which makes it take

root in memory in a way which would be otherwise impossible.

To me the functional possibilities of timbre only seem valid if they are linked to language and to the articulation of a discourse through structural relationships; timbre both explains and masks at the same time. Without musical discourse it is nothing, but it can also form the entire discourse on its own.

I have dwelt so long on the question of orchestral writing since I believe that much can be learnt from it. I very much deplore the wide gap which exists between composers who have learnt to write for the orchestra and know how to use it, and composers who exclusively use new electro-acoustic means. I am sure that the latter would avoid a certain amount of tentative stumblings and set-backs if they knew how to transpose all the accumulated orchestral technique into other areas; if they could recognize the principles which have guided the evolution of orchestral timbre, and approach the use of synthetic timbre by drawing from the rich store that this evolution has brought, the provisional result of which is infinitely varied and extremely rich. What often worries me when I hear works based on sound synthesis is either the confusion between levels of composition, or the great concentration on sound-objects and the lack of concern regarding the relationships between them. Sometimes these sound-objects are very beautiful and well thought of, but they possess a very vigorous centrifugal force and so have great difficulty integrating themselves into a real discourse. They have a tendency, rather, to group themselves in a collage that is neither very subtle nor very solid. Differentiation in the levels of perception and variation in the use of time with regard to its density or specificity can be some of the functions of timbre in the articulation of forms.

A major, frequently mentioned preoccupation with regard to invented timbres is continuity. I believe, without paradox, that the notion of discontinuity is far more important both in the fields of timbre and of pitch. Discontinuity provides a means of transition and of this frequently-mentioned continuity, as much as it provides abrupt divisions and separations. In short, discontinuity gives rise to composition in its many dimensions.

The notions of polyphony and heterophony would need to be re-thought and re-evaluated in a completely different context: true polyphony, where structures are deduced one from another, following more or less strict laws; heterophony where structures based on similar principles create an apparent divergence. Such notions remain fundamental. They need to be re-assessed in the context of a new sound-world which pre-supposes a more *quantified* means of investigation than before.

Finally, to conclude this brief survey of timbre, I shall express a wish: that the transmission of sound be studied very seriously, and that a radical renewal should be attempted in this field. Today's loudspeaker is a great anonymous pulveriser of sound that does not measure up to the means which have been developed to create a new sonic world. In an orchestra the transmission of the sound of instruments is infinitely

varied, depending on their shape, their power, their ability to radiate sound. The loudspeaker discharges sound through its dark sterile mouth: if it has the advantage of being able to *relocate* sound at will, it has the grave disadvantage of causing a *reduction* in the sound. I saw a play in London where the actors in the first part spoke without amplification; in the second they all had their voices amplified. It was as easy to recognize their "natural" voices as it was difficult to hear the amplified voices which were lost in an anonymous fog. This wasn't through a lack of quality in transmission, but because the loudspeaker forcibly compressed the register of the voices into a formidable uniformity. The loudspeaker is not so much a *polyvalent* tool as a piece of *non-valent* equipment, if I may say so. The projection of sound will remain problematical as long as the relationship between sound transmission and timbre is not carefully studied. Timbre and its transmission are characteristics which I judge to be inseparable one from the other.

To briefly provide a provisional conclusion, I would underline that timbre, through composition, should integrate itself totally with musical language in a multidimensional world where its specificity will be the measure of its importance.

Notes

1. This paper was first presented on April 17, 1985 at the Seminar on Timbre held at IRCAM, Paris, France.
2. The term "musical composition" which appears several times in this text does not fully convey the meaning of the French word "*écriture*" which has implications of symbolic reasoning. (Cf. Bonnet article, this volume, footnote 1).

Translated by R. Robertson